

DIONIS LUDOVICK NASUWA

Full-Stack Software Engineer & Systems Developer

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PROFESSIONAL SUMMARY

Computer Science student at the **University of Dar es Salaam (CoICT)** with strong competencies in full-stack web engineering, distributed systems architecture, and database design. Certified in **Full Stack Open (University of Helsinki, Grade 5/5)** and **Scrum Fundamentals (SFC™)**. Experienced in developing responsive web applications, RESTful APIs, and distributed monorepos using TypeScript, JavaScript, Python, Node.js, and React, backed by practical experience in network infrastructure optimization.

EDUCATION & VERIFIED CERTIFICATIONS

BSc in Computer Science — University of Dar es Salaam (CoICT) Expected Oct 2026
Full Stack Open Certification — University of Helsinki (Grade 5/5 Maximum Score) July 2026
Scrum Fundamentals Certified (SFC™) — SCRUMstudy Professional Certification July 2026
Key Coursework: Data Structures & Algorithms, Database Systems (SQL), Object-Oriented Analysis & Design (OOAD), Operating Systems & Linux Admin, Computer Networks, Systems Programming.

TECHNICAL EXPERIENCE

Software Engineering Collaborator • Mainstream Group (Technology Division) March 2026 – May 2026 | Dar es Salaam

- Architected modular application features using **TypeScript** and **Node.js** across multi-service business applications, improving data throughput and system interoperability.
- Built secure **RESTful API endpoints** and backend data pipelines, participating in bi-weekly Agile sprint workflows, code reviews, and milestone delivery.

Software Development Trainee • UDICTI / StartHub Africa Partnership July 2025 – Sept 2025 | Dar es Salaam

- Engineered cross-platform mobile interfaces and backend integration for the **Smart Cargo** logistics platform using **Flutter** and **Django**.
- Implemented dynamic multi-variable rate calculations (distance, weight, fragility) with **OpenStreetMap** API and localized Swahili/English UI modules.

IT Infrastructure Trainee • TPC Limited July 2024 – Sept 2024 | Kilimanjaro

- Configured managed network switches, deployed routers, and set static IP bindings to optimize enterprise LAN uptime and resolve hardware/OS diagnostics across 50+ workstations.

SELECTED TECHNICAL PROJECTS

Plaque Platform — Digital Identity Monorepo TypeScript · Next.js · Fastify · Turborepo · Prisma · Docker

- Architected an 8-app distributed monorepo with **Turborepo** remote caching, integrating an asymmetric **RS256 JWT** authentication microservice for zero-latency cryptographic verification.
- Configured isolated **Prisma ORM** schemas per service on PostgreSQL with containerized **Docker** deployments.

INTLEQ — Local LLM RAG & Vector Search System Python · FastAPI · Ollama · pgvector · PostgreSQL · Docker

- Built an offline-first Retrieval-Augmented Generation (RAG) platform executing local inference via **Ollama (llama3.2)** without cloud API dependencies.
- Engineered an asynchronous **FastAPI** backend querying **pgvector** cosine similarity embeddings with containerized PostgreSQL.

Strata (Final Year Project) — AST Code Analysis Engine Python · AST Parsing · Graph Theory (SCC) · Streamlit · Docker

- Engineered an analytical code engine converting source code into **Abstract Syntax Trees (ASTs)** and **Directed Acyclic Graphs (DAGs)**.
- Implemented **Strongly Connected Components (SCC)** clustering algorithms and interactive **Streamlit** dashboards to isolate circular code entanglements.

TECHNICAL SKILLS & COMPETENCIES

Languages: JavaScript (ES6+), TypeScript, Python, SQL, Dart

Frontend & Mobile: React, Next.js (App Router), Flutter, Redux Toolkit, Tailwind CSS, Vite, Streamlit

Backend & Databases: Node.js, Express.js, Fastify, FastAPI, Django, PostgreSQL, MySQL, Redis, Supabase, pgvector, Prisma ORM, RESTful APIs

DevOps & Infrastructure: Docker, Turborepo, Git/GitHub Actions, Linux (Ubuntu/Fedora), Nginx, Static IP Routing, LAN Troubleshooting

Core Methodologies: Agile / Scrum (SFC™ Certified), Object-Oriented Programming (OOP), Test-Driven Development (TDD), AST & Graph Analysis